

SHIVAM SARASWAT

Senior Product Security Engineer

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PROFESSIONAL SUMMARY

Senior Product Security Engineer with 4+ years of experience architecting enterprise-grade security solutions, securing CI/CD pipelines, and delivering cloud-native security solutions at scale. Deep expertise in SSDLC automation, container security, vulnerability management, supply chain security, and developer-centric tooling. Known for reducing vulnerability noise, security risk, and alert fatigue, improving remediation speed, streamlining compliance processes and shipping security tooling that developers actually use. Committed to bridging the gap between development and security to create resilient, scalable, and efficient systems.

EXPERIENCE

Senior Cybersecurity Engineer

PayPal

April 2025 – Ongoing Bengaluru, Karnataka

- Engineered Org-Wide Supply Chain Firewall:** Developed a multi-stage package security gateway upstream of Artifactory – enforcing age-based blocking, malicious package database checks (OSV.dev), and vulnerability scanning before any package reaches Artifactory for developer consumption.
- Architected Container Security Framework:** Standardized vulnerability policy enforcement across all engineering teams, achieving 100% compliance and accelerating remediation cycles organization-wide.
- CI/CD-Native Security Enforcement:** Deployed a centralized container security integration across 3,100+ pipelines to auto-block non-compliant images, improving vulnerability detection by 50%.
- Benchmarked Container Scanning Tools:** Evaluated detection accuracy, performance, cost, and API support – piloting the best-fit solution within pipeline SLA constraints.
- Architected End-to-End Third-Party Vendor Image Scanning Framework:** Designed fully automated CI/CD pipeline and standardized onboarding workflows – including a Slack AI bot enabling developers to trigger scans on-demand via chat.
- Built Intelligent Vulnerability Deduplication System:** Engineered CVE-, package-, and version-based deduplication solution across Container and SCA scanners with image layer-based analysis – reducing alert fatigue and improving patch efficiency by 30%.
- Engineered AI-Powered SCA Intelligence System:** Built AI-powered prototype for reachability analysis to eliminate false positives, safe-version recommendations to prevent breaking changes, and intelligent parent dependency fix suggestions across Maven and npm – improving remediation accuracy.
- Supply Chain Attack Prevention:** Identified and neutralized the Shai-Hulud self-replicating worm and multiple ongoing similar supply chain campaigns, preventing credential theft and lateral spread across build environments.
- Led Cross-Language Vulnerability Research:** On Go CVEs surfacing in Node.js runtimes and build-time vs. runtime exploitability – sharpening risk assessment and remediation prioritization.
- Developed Comprehensive Container Security Documentation Hub:** Authored ADRs, integration guides and workflow docs reducing developer onboarding time by 40%.

EDUCATION

B.Tech in Computer Science and Engineering (Cyber Security)

GLA University, Mathura

May 2022 8.25 CGPA

SKILLS

Security Expertise:

- AppSec & DevSecOps:** SSDLC, Container Security, Application Security, Cloud Security, Vulnerability Management, Policy as Code, Supply Chain Security, OWASP Top 10, Threat Modeling, Secure Code Review
- Vulnerability Research:** Security Research, Root Cause Analysis (RCA), Vulnerability Triage & Prioritization

Security Tools & Platforms:

- SAST:** Semgrep, CodeQL (GHAS), Bandit
- SCA & Dependency Management:** Dependabot, Renovatebot, Syft (SBOM), Dependency-Track
- Secret Scanning:** Gitleaks, TruffleHog, GitHub Advanced Security, Talisman
- Container Security:** Wiz, JFrog Xray, Trivy, Hadolint
- DAST & Security Testing:** Nuclei, OWASP ZAP, Burp Suite Professional
- Network & Vulnerability Scanning:** Nmap, Nessus, Wireshark
- Monitoring & Observability:** Elasticsearch, New Relic, Datadog

Core Technical Skills:

- Programming & Scripting:** Python, Bash, PowerShell, Git
- Cloud & Infrastructure:** Google Cloud, AWS, Docker, Kubernetes (K8s)
- CI/CD & DevOps:** GitHub Actions, GitLab CI/CD, Harness CI/CD
- Databases & APIs:** MongoDB, REST APIs, Swagger, Postman

Security Engineer

Tekion

📅 August 2024 – March 2025 📍 Bengaluru, Karnataka

- **Architecting DevSecOps Pipelines:** Architected and deployed comprehensive DevSecOps security scanning pipelines — SAST, SCA, Secret Scan, Container Scan, and Dockerfile Scan — across 2,000+ production repositories, achieving a 40% reduction in security vulnerabilities.
- **DevSecOps Pipelines Migration:** Led the migration of 2,000+ production repositories to GitLab's Pipeline Execution Policy, ensuring seamless transition with zero production disruption.
- **Pipeline Optimization & Developer Collaboration:** Optimized pipeline execution time by 35% through workflow and configuration tuning, while refining templates based on developer feedback to improve overall developer experience.

Product Security Engineer

IKEA

📅 April 2023 – August 2024 📍 Bengaluru, Karnataka

- **Architecting Security Solution & Deployment:** Engineered in-house Automated Web and API Security Monitoring Solution, reducing external engagement costs by 20% and improving the responsible disclosure program's response capability.
- **SSDLC Automation & Integration:** Architected "Argos", an org-wide proactive SSDLC maturity automation platform offering one-click security enablement, on-demand security assessment, maturity scoring, actionable insights, and secure template integration for developers.
- **Automated Cloud Vulnerability Management & Visualization:** Built an end-to-end triage and notification system with centralized dashboard for Google Security Command Center (SCC), cutting critical-issue response time by 40%.
- **Shadow IT Asset Management:** Resolved numerous DNS dangling issues associated with shadow IT assets, preventing potential subdomain takeovers.
- **Managed Bug Bounty Program:** Investigated and resolved multiple issues reported by external Security Researchers.
- **Security Awareness:** Delivered 10+ engaging sessions with actionable cybersecurity strategies to non-security co-workers.

Security Automation Engineer

BreachLock

📅 March 2022 - April 2023 📍 Noida, UP

- **Cyber Threat Research & Automation Code:** Researched emerging threats and developed automation code for Vulnerability Scanners and an External Attack Surface Management platform.
- **Vulnerability Scanner Enhancement:** Improved scanner effectiveness by incorporating vulnerabilities discovered during penetration tests into detection logic and workflows.
- **Security Automation & Testing:** Developed modular security automation plugins with documentation and Pytest-based test coverage to improve reliability, scalability, and maintainability.

ACCOMPLISHMENTS



Best Security Product of the Year
2nd Annual Cyber Security Excellence Awards 2023 by Quantic India



Hackathon Runner Up
IKEA's InnerSource Hackathon 2024



Speaker at BSides Vizag 2025
Presented – "Breaking Bad: Container Security is Broken" — covering layer-aware vulnerability analysis and threat intelligence (CISA KEV, EPSS) to surface vulnerabilities that actually matter



Top 1% (Legend Rank) on TryHackMe
Completed 120+ rooms on DevSecOps, Pentesting, CTFs, OWASP Top 10, etc.

PERSONAL PROJECTS



SeCoRA (Secure Code Review AI Agent)
An AI-powered tool for detecting and remediating security vulnerabilities in codebases

PUBLICATIONS



Conference Proceedings
• A. Tiwari, S. Saraswat, U. Dixit, and S. Pandey, "Refinements in zeek intrusion detection system," in *2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)*, vol. 1, 2022, pp. 974–979. DOI: 10.1109/ICACCS54159.2022.9785047.

CERTIFICATIONS

- **EC-Council** - Certified Ethical Hacker (Practical)
- **IBM** - Cyber Security Graduate
- **PentesterLab** - Unix Badge
- **Fortinet** - NSE 1 Network Security Associate

LANGUAGES

English
Hindi

